AIML lab - Assignment 3 21-08-2026

1. Accept 10 random integers into a list and display their sum, min and max.

```
numbers = list(map(int, input("Enter 10 integers: ").split()))

print("Sum:", sum(numbers))
print("Minimum:", min(numbers))
print("Maximum:", max(numbers))
```

Output:

```
Enter 10 integers: 12 5 8 20 3 15 7 10 6 18
Sum: 104
Minimum: 3
Maximum: 20
```

2. Accept empno, ename, and basic salary into a dictionary. Calculate da as 5% of basic, hra as 7% of basic, and net as basic + da + hra, and store them in the dictionary. Display all the details from the dictionary using its keys. Display all the keys and all the values of the dictionary using appropriate dictionary functions

```
emp = {}

emp["empno"] = int(input("Enter employee number: "))
emp["ename"] = input("Enter employee name: ")
emp["basic"] = float(input("Enter basic salary: "))

emp["da"] = emp["basic"] * 0.05
emp["hra"] = emp["basic"] * 0.07
emp["net"] = emp["basic"] + emp["da"] + emp["hra"]

print("\nEmployee Details:")
for key in emp:
    print(key, ":", emp[key])

print("\nAll Keys:", emp.keys())
print("All Values:", emp.values())
```

Output:

```
Enter employee number: 101
Enter employee name: Ravi
Enter basic salary: 20000

Employee Details:
empno : 101
ename : Ravi
basic : 20000.0
da : 1000.0
hra : 1400.0
net : 22400.0

All Keys: dict_keys(['empno', 'ename', 'basic', 'da', 'hra', 'net'])
All Values: dict_values([101, 'Ravi', 20000.0, 1000.0, 1400.0, 22400.0])
```

3. Accept values for two sets and display if the second set is a subset of the first set.

```
set1 = set(map(int, input("Enter first set: ").split()))
set2 = set(map(int, input("Enter second set: ").split()))

if set2.issubset(set1):
   print("Second set is a subset of first set")
else:
   print("Second set is not a subset of first set")
```

Output:

```
Enter first set: 1 2 3 4 5
Enter second set: 2 4
Second set is a subset of first set
```

4. Copy assignment 2 to a text file and name the file as document.txt. Write a python program to read the file and display the maximum occurring word in the file and its count and the minimum occurring word in the file and its count.

```
from collections import Counter

with open("document.txt", "r") as file:
    words = file.read().split()

count = Counter(words)

max_word = max(count, key=count.get)
min_word = min(count, key=count.get)

print("Maximum occurring word:", max_word, "Count:", count[max_word])
print("Minimum occurring word:", min_word, "Count:", count[min_word])
```

document.txt:

Python is easy Python is powerful Python

Output:

```
Maximum occurring word: Python Count: 3
Minimum occurring word: easy Count: 1
```

5. Populate a 3 X 3 matrix with random numbers and display its transpose.

```
import random

matrix = [[random.randint(1, 9) for j in range(3)] for i in range(3)]

print("Matrix:")
for row in matrix:
    print(row)

print("Transpose:")
```

```
for j in range(3):
    for i in range(3):
        print(matrix[i][j], end=" ")
    print()
```

Output:

```
Matrix:
[2, 5, 8]
[1, 7, 3]
[6, 4, 9]

Transpose:
2 1 6
5 7 4
8 3 9
```

6. Populate one 3 X 3 matrix with random numbers. Populate another 3 X 3 matrix with the value 25. Accept a choice to add, subtract and matrix multiplication of the two matrices. Finally display the two matrices and the output matrix depending on the choice accepted..

```
import random

A = [[random.randint(1, 9) for j in range(3)] for i in range(3)]
B = [[25 for j in range(3)] for i in range(3)]

print("Matrix A:")
for row in A:
    print(row)

print("Matrix B:")
for row in B:
    print(row)

choice = int(input("Enter choice (1-Add, 2-Subtract, 3-Multiply): "))

if choice == 1:
    C = [[A[i][j] + B[i][j] for j in range(3)] for i in range(3)]
```

```
elif choice == 2:
    C = [[A[i][j] - B[i][j] for j in range(3)] for i in range(3)]
elif choice == 3:
    C = [[sum(A[i][k] * B[k][j] for k in range(3)) for j in range(3)] for i in range(3)]
else:
    print("Invalid choice")
    exit()

print("Output Matrix:")
for row in C:
    print(row)
```

Output:

```
Matrix A:
[2, 5, 8]
[1, 7, 3]
[6, 4, 9]

Matrix B:
[25, 25, 25]
[25, 25, 25]
[25, 25, 25]

Enter choice (1-Add, 2-Subtract, 3-Multiply): 1

Output Matrix:
[27, 30, 33]
[26, 32, 28]
[31, 29, 34]
```

7. Accept sentences through the keyboard and add them to a file greet.txt while ans=”yes”. Later accept a sentence and check if the sentence exists in the file.

```
while True:
    sentence = input("Enter a sentence: ")

    with open("greet.txt", "a") as file:
```

```
        file.write(sentence + "\n")

    ans = input("Do you want to continue? (yes/no): ")
    if ans.lower() != "yes":
        break

search = input("Enter sentence to search: ")

with open("greet.txt", "r") as file:
    sentences = file.read().splitlines()

if search in sentences:
    print("Sentence exists in the file")
else:
    print("Sentence does not exist in the file")
```

Output:

```
Enter a sentence: Hello Python
Do you want to continue? (yes/no): yes
Enter a sentence: Python is easy
Do you want to continue? (yes/no): no
Enter sentence to search: Python is easy
Sentence exists in the file
```